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PVM & Verification

Owens: Verification engine (*DOM→semantic→visual*), IndexedDB + LRU memory, state-hash matching, recovery loop.

Role 5 of 6 · PVM

HW Hardware access

None. This is browser-storage and logic work end to end.

No booking needed — develop and test on any laptop, including the lowest-spec one on the team.

P0 Phase 0 — now → 20 Sep 2026 (lock the idea)

- Design the memory key (session + tab + page/state hash + task context) together with Perception, and the IndexedDB schema.
- Write down explicitly what PVM must **never** store (raw screenshots, passwords, PII, unverified transitions) — Integration can audit against this list later.

P1 Phase 1 — build the MVP (20 Sep → shortlist)

- Hierarchical verification: Level 1 deterministic (URL/DOM/events) → Level 2 semantic (success/error text) → Level 3 visual comparison only when ambiguous.
- Recovery loop: failed/ambiguous action → capture new state → re-evaluate → request server reasoning if needed → retry → verify → update memory.
- Predict → Validate → Execute → Verify → Memorize loop, with bounded IndexedDB storage and LRU eviction.

P2 Phase 2 — rehearse & harden (shortlist → Dec finale)

- Build the deliberate “stale cache” demo with Integration: change the UI, show PVM detect the mismatch and fall back correctly — the first thing a skeptical judge will try to break.

DEP Dependencies

YOU DEPEND ON	Extension (action execution + result state to verify), Server AI (the first verified action to memorize), Perception (state-hashing format).
DEPENDS ON YOU	Integration (benchmark harness needs your hit/miss + latency-delta data), Extension (your recovery loop hooks back into its execution path).

Full team plan, hardware-booking rules, and the master phase timeline: see the companion *PS26171 Team Workflow & Hardware Plan* document.